

$$1. f(x, y) = 2x^2 - xy - 3y^2 - 3x + 7y$$

$$\frac{\partial f}{\partial x} = 4x - y - 3 = 0$$

$$\frac{\partial f}{\partial y} = -x - 6y + 7 = 0$$

$$\begin{aligned} 4x - y - 3 &= 0 \\ 4 \cdot -x + 6y + 7 &= 0 \end{aligned}$$

$$\begin{aligned} x &= 1 \\ y &= 1 \end{aligned} \quad (1, 1)$$

$$25y + 25 = 0$$

$$\frac{\partial^2 f}{\partial x^2} = 4$$

$$\frac{\partial^2 f}{\partial y^2} = -6$$

$$\frac{\partial^2 f}{\partial x \partial y} = -1$$

$$D = f_{xx}f_{yy} - (f_{xy})^2 = -25 \quad D < 0$$

This is a saddle point.

$$3. A^T A x = A^T b$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3/2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3/2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 15/2 \\ 15/2 & 21 1/4 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3/2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 + 2 + 1 \\ 2(1) + 3/2(2) + 4(1) \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 15/2 \\ 15/2 & 89/4 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$$

get rid of fractions

$$\begin{bmatrix} 12 & 30 \\ 30 & 89 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 16 \\ 36 \end{bmatrix}$$

use Cramer's

$$x = M^{-1} y$$

$$M^{-1} = \frac{1}{168} \begin{bmatrix} 89 & -30 \\ -30 & 12 \end{bmatrix}$$

$$M^{-1} y = \frac{1}{168} \begin{bmatrix} 89 & -30 \\ -30 & 12 \end{bmatrix} \begin{bmatrix} 16 \\ 36 \end{bmatrix} = \begin{bmatrix} 43/21 \\ -2/7 \end{bmatrix} = \begin{bmatrix} c \\ d \end{bmatrix}$$

4. Linear interpolation formula:

$$f(x) \approx f(x_0) + \frac{f(x_1) - f(x_0)}{(x_1 - x_0)} (x - x_0)$$

given

$$f(0.25) \approx 1 + \frac{1.6487 - 1}{0.5 - 0} (0.25 - 0)$$
$$\boxed{\approx 1.32435}$$

5. Using Central Difference:

$$v(t_i) \approx \frac{x_{i+1} - x_{i-1}}{2h}$$

@ $t = 0.5$

$$v(0.5) \approx \frac{6.8 - 0}{2 \cdot 0.5} = \frac{6.8}{1} = \boxed{6.8 \text{ m/s}}$$

@ $t = 1.25$

$$v(1.25) \approx \frac{9.9 - 6.8}{1} = \boxed{3.10 \text{ m/s}}$$

Using Mid point Formula:

$$a(t_i) \approx \frac{x_{i+1} - 2x_i + x_{i-1}}{h^2}$$

@ $t = 1.5$

$$a(1.5) \approx \frac{12.15 - 2(9.90) + 6.80}{(0.5)^2} = \boxed{-3.4 \text{ m/s}^2}$$

6. Simpson's:

$$\int_0^2 3x^2 dx \approx \frac{1}{3} [0 + 4(3) + 12] = \boxed{8}$$

Trapezoid:

$$\int_0^2 3x^2 dx \approx \frac{1}{2} [0 + 2(3) + 12] = \boxed{9}$$